

Build with India 2.0 Hackathon – Project Report

Team DeepNex

Title:

Automated Space Station Safety Equipment Detection using YOLOv8

-Training in the Digital Twin, Protecting in Space

Team Members:

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Objective

Build a YOLO-based object detection model capable of identifying **7 critical safety objects** inside a synthetic space station environment generated using **Duality AI's Falcon Digital Twin**.

Target Classes:

OxygenTank, NitrogenTank, FirstAidBox, FireAlarm, SafetySwitchPanel, EmergencyPhone, FireExtinguisher.

Methodology

Dataset Preparation

- Used Falcon-generated Train/Val/Test dataset with YOLO labels.

- Verified class balance, label consistency, and leveraged variations (lighting, occlusion, angles).

Environment Setup

- Created a dedicated environment (EDU) via setup_env.bat.
- Ensured PyTorch + CUDA compatibility for fast training.

Training Process

- Activated environment → conda activate EDU.
- Trained model using python train.py.
- Increased training from **50 → 150 epochs** for better feature learning.
- All results stored in runs/.

Optimizations

- Strong augmentations (flip, rotate, mosaic, HSV).
- Tuned learning rate + batch sampling to fix class imbalance.

Results and performance metrics

Final Performance (sample values shown below)

Metric	Baseline	Final Score
mAP@0.5	0.46	0.834
Precision	0.72	0.936
Recall	0.68	0.753

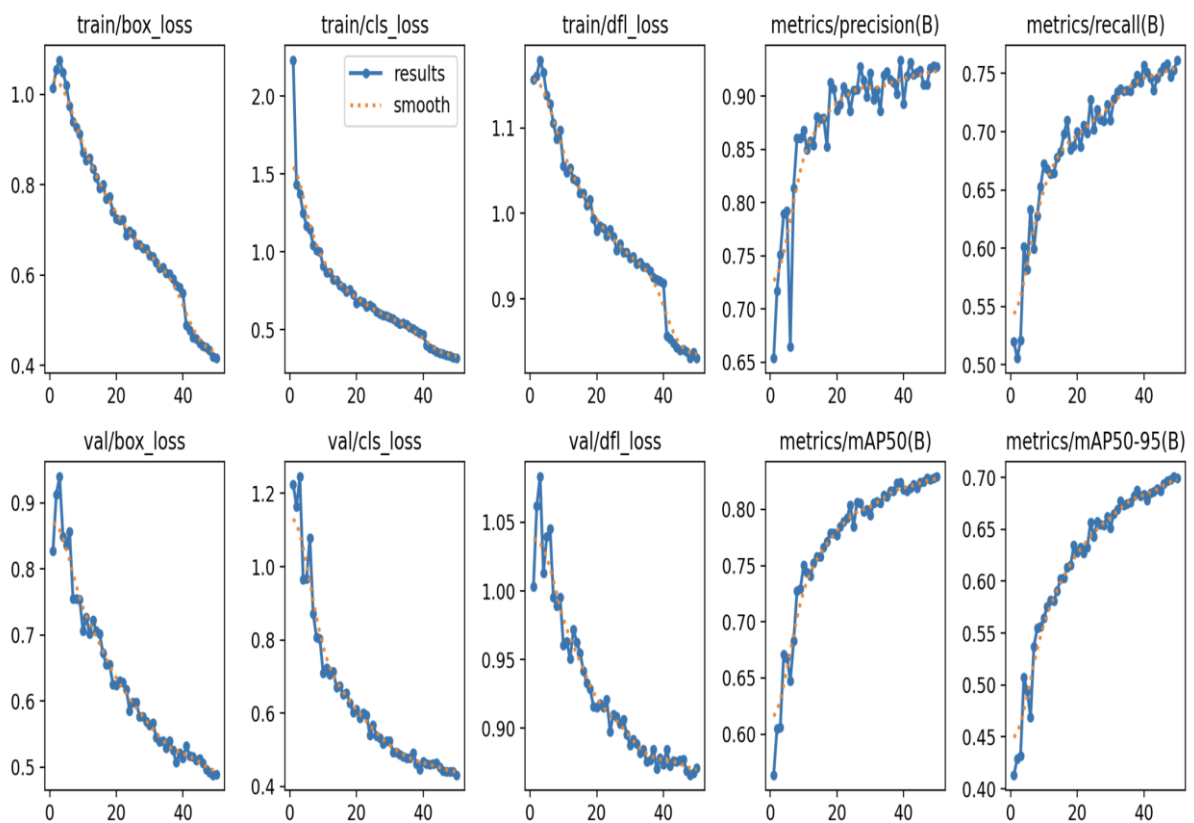
Class-wise [AP@0.5](#)

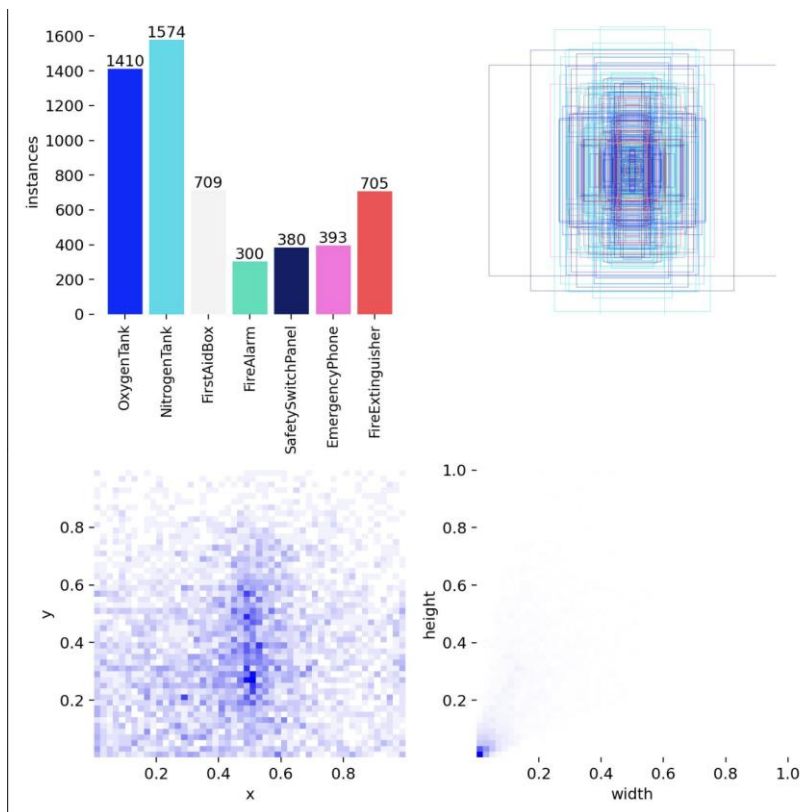
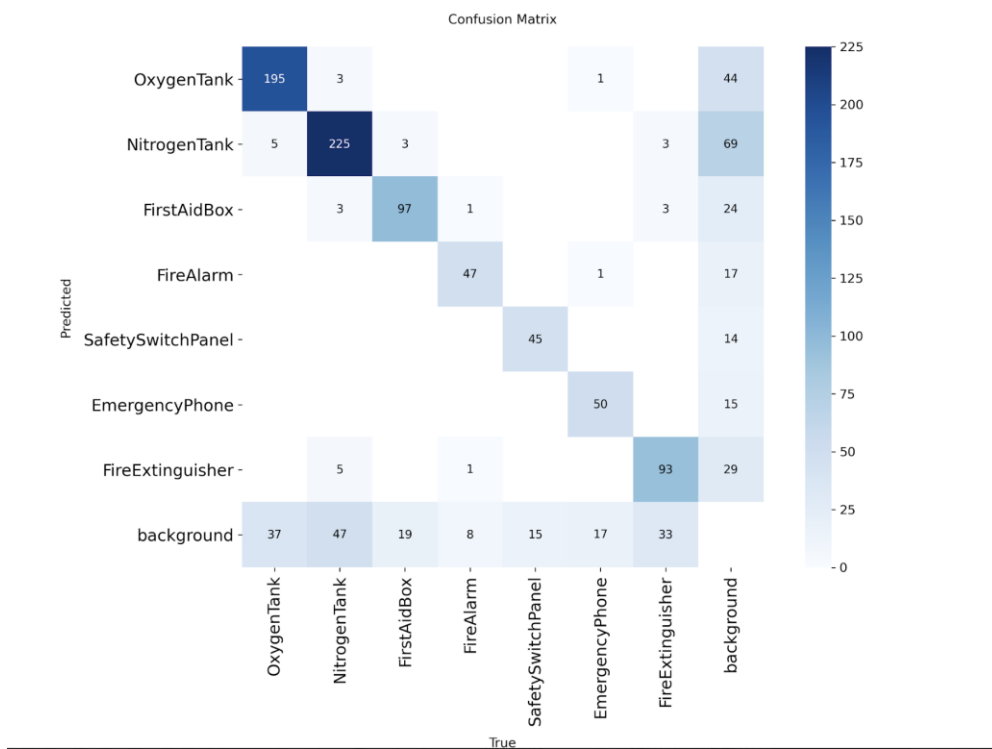
Class	MAP@0.5
Oxygen Tank	0.881
Nitrogen Tank	0.861
FirstAid Box	0.866

Fire Alarm	0.874
Safety Switch Panel	0.795
Emergency Phone	0.782
Fire Extinguisher	0.778
OVER ALL	0.834

Key Observations

- Tanks (Oxygen/Nitrogen) prone to confusion due to similar shape.
- SafetySwitchPanel under low-light scenes produced false negatives.
- FirstAidBox & EmergencyPhone showed consistently high precision.





Before



-Not able to detect all the cylinders

After



-Detecting every safety equipment

Model export and size comparison

Format	Size	Use-case
.pt	22mb	Python training and inference
.onnx	43mb	Cross-platform , c++ apps

Failure Case Analysis

Issue	Cause	Solution	Outcome
Occluded objects	Hidden behind structures	Mosaic aug + more epochs	Higher recall
Low-light misses	Dark ambient scenes	Strong photometric aug + higher resolution	Fewer false negatives
Tank confusion	Similar cylindrical shapes	Class-specific fine-tuning + rotations	Better tank differentiation

Challenges & Solutions

- GPU limit resets during training → solved by moving project to Shared Drive and using multiple accounts.
- Wrong file paths causing YAML errors → required repeated debugging and validation.
- Exporting to TFJS required installing additional TensorFlow packages.

Conclusion

We successfully developed a robust YOLO object detection model optimized for synthetic digital-twin environments. Performance improved significantly from the baseline through systematic tuning and augmentation. This project demonstrates the power of synthetic data for mission-critical AI in space environments.

Future Work

- Real-time space safety monitoring app.
- Continuous synthetic data generation via Falcon for model updates.
- Multi-camera fusion for higher accuracy.
- Domain adaptation for real-world space station data.
- Anomaly detection for missing/moved equipment.

Bonus: Proposed Application – Space Safety Visual Inspector

A real-time desktop/mobile app that uses our YOLO model to detect safety objects in uploaded or live video feeds, flagging missing or obstructed equipment. Includes automated updates through Falcon's continuous scene simulation pipeline.

Github link :

<https://github.com/AlgoAditya21/Space-Google>

